HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78AA

ACTUARIAL AND FINANCIAL MATHEMATICS A

Semester 1 2019/20

Duration: Two Hours

Total Marks: 60

ANSWER ALL 8 QUESTIONS

Electronic calculators approved by the University may be used.

1. Give your answers rounded to 6 decimal places in the following questions.

(a) Given that the effective rate of discount is 0.5% per month, calculate the effective rate of interest per annum.

[2 marks]

(b) The effective rate of interest is 0.5% per month. Calculate the effective rate of discount per half-year.

[2 marks]

[**Total 4 marks**]

2. Find the present value at outset of a level annuity of £1000 per year, payable half-yearly in arrears for 10 years, deferred for 12 years, if the effective rate of interest is 10.25% per annum during the first 16 years and 4% per half-year thereafter.

[**Total 4 marks**]

3. A series of cashflows is given as follows:

Date	**Cashflow**
1 June 2019	+100
1 February 2020	-85
1 October 2021	-25

The effective rates of interest are 4% per annum in 2019, 2.5% per quarter in 2020 and 1% per month thereafter.

What is the net accumulation of these cashflows on 1 January 2022?

[**Total 4 marks**]

4. A bond currently available in the market has the following terms:

- Remaining term : 7.5 years.
- Coupon : 8% per annum payable twice a year in arrear.
- Redemption : At a discount of 2%.

The coupon due now has just been paid. An investor subject to income tax at 25% and capital gains tax at 40% purchased £100 nominal of this bond at £65.85. Calculate the net redemption yield per annum he would obtain from this bond.

[**Total 6 marks**]

5. The following table gives information concerning an investment fund:

Date	Fund Value	Cashflow IN	Cashflow OUT
1 January 2016	100		
1 February 2016	101	1	
1 June 2016	103		
1 January 2017	106		2.5
1 June 2017	104		1
1 December 2017	105	5	
1 January 2018	112		

Assumptions:
All values are in millions.
All cashflows occur immediately after the fund value is given.

(a) Calculate the annual Money Weighted Rate of Return for the above fund.

[5 marks]

(b) Calculate the annual Time Weighted Rate of Return for the above fund.

[4 marks]
[**Total 9 marks**]

6. An investor who has £5 million of her own money to invest wants to purchase an apartment block for £6 million. She can borrow the shortfall of £1 million from a bank at the rate of 0.5% per month effective. Additionally, she receives a rate of 0.4% per month on any deposits she makes.

The investor believes that she can get rental income of £25,000 per month payable at the end of each month for the next 5 years at which point she can sell the apartment block for £7 million.

(a) Find the discounted payback period for the £1 million loan if she uses all the rental income she receives to repay the loan as soon as possible.

[5 marks]

(b) Find the accumulated profit at the end of the 5 years, assuming she uses all the income associated with this investment to repay the loan as soon as possible.

[5 marks]
[**Total 10 marks**]

7. On 1 January 2010, a student takes out a 10-year bank loan for £50,000 that will be repaid by instalments at the end of each month. Under this repayment schedule, the instalments at the end of each month in 2010 will be X, the instalments at the end of each month in 2011 will be 2X and so on, until the instalments at the end of each month in 2019 will be 10X. The bank charges a rate of interest of 9% per annum effective.

(a) Show, using the Principle of Equivalance, that

$$50000 = X\left[a^{\frac{j}{}}_{\overline{m}|}\right]\left[(I\ddot{a})^{\frac{i}{}}_{\overline{n}|}\right]$$

where, j = the effective rate of interest per month,
i = the effective rate of interest per year,
m and n are integers that need to be determined.

[5 marks]

(b) Hence show that X = £130, rounded to the nearest pound.

[3 marks]
[**Total 8 marks**]

8. Joel takes out a mortgage loan of £500,000 which is repayable by an annuity payable annually in arrears for 30 years. After the first annual repayment, each subsequent repayment increases by 5%. The annual repayments were calculated at an effective rate of interest of 10% per annum.

(a) Show that the first annual repayment amount is £33,230.73 (minor errors due to rounding will not be penalised).

[5 marks]

(b) Immediately after the fifteenth repayment, the interest rate on the outstanding loan is increased to 12% per annum effective. Calculate the amount of the sixteenth repayment instalment if subsequent repayment instalments are increased by £2,500 each year, and the loan is to be repaid by the original date, i.e. 30 years from commencement.

[10 marks]
[**Total 15 marks**]

END OF PAPER